

IBPS SO Previous Year Paper 2021

SECTION I: ENGLISH LANGUAGE (Q. 1–50)

Directions (1–7): Read the following passage and answer the questions. Some words are highlighted to help you answer some of the questions.

India's demographic landscape is undergoing a dramatic transformation. The country, which is now the most populous nation on earth, faces the paradox of a youthful population and an ageing workforce simultaneously in different regions. Urban centres are witnessing an unprecedented influx of migrants, straining civic infrastructure, while several rural belts grapple with depopulation. The consequences are **disconcerting (A)**. These visible symptoms, however, are mere expressions of a deeper structural _____(B)_____ that the nation must urgently address. The vitality of any society rests on the combined output of its human capital and institutional frameworks. **(C)** Robust **invest (1)** in education and skill development form the **cornerstone (2)** of such socio-economic **progress (3)** among citizens who **trust (4)** national growth. Our social compact of aspiration and opportunity is now fraying at the edges.

There is a palpable sense of disillusionment among the youth. Many first-generation graduates find themselves in jobs that do not match their qualifications or aspirations. The gig economy, while offering flexibility, provides little by way of social security or long-term stability. Formal sector employment has stagnated, and entrepreneurship faces steep regulatory and financial barriers. The technology sector, an engine of high-value employment and innovation, _____(D)_____. Policymakers and academics are increasingly hesitant to acknowledge systemic design flaws, for fear of political repercussions. There is a profound sense of institutional inertia that dampens economic dynamism. When ambition among citizens is suppressed, it erodes the productivity and innovation cycle, and eventually, _____(E)_____. This pervasive climate of stagnation and unfulfilled potential is a fundamental drag on inclusive economic progress.

(F) Confidence in independent arbiters, such as the judiciary, the free press, regulatory watchdogs and academic institutions, has been progressively hollowed out. With the erosion of institutional credibility, there is no reliable recourse for citizens seeking protection from arbitrary policy changes or market failures. **(G)** This causes **entrepreneurs (1)** to curtail their risk **undertaking (2)** even further, deterring **appetite (3)** from investing in new ventures and **creating (4)** jobs. This toxic combination of institutional distrust, regulatory uncertainty and dimming aspirations is slowly but surely strangling India's demographic dividend.

Q1. Which of the following options represents a pair of synonym and antonym respectively for the word marked as **(A)** in the given passage?

- | | |
|--------------------------|--------------------------|
| (a) Placate, Inflamm | (b) Alarming, Reassuring |
| (c) Invigorating, Soothe | (d) Perturbing, Mollify |
| (e) None of these | |

Q2. Which of the following words given in the options should come at the place marked as **(B)** in the above passage? The word should also fit in the two sentences given below:

- (I) The investigation revealed a deep-seated institutional _____ that had persisted for decades without corrective intervention.
(II) Analysts pointed to a fundamental _____ in the regulatory framework as the root cause of the repeated market collapses.

- | | |
|-------------------|-------------|
| (a) Benevolence | (b) Inertia |
| (c) Malaise | (d) Ardour |
| (e) None of these | |

Q3. The sentence given in **(C)** has four words given in bold. Amongst the given bold words, which must interchange to make the sentence grammatically and contextually correct?

- | | |
|-------------|-------------|
| (a) (2)–(4) | (b) (1)–(2) |
| (c) (3)–(4) | (d) (1)–(4) |
| (e) (2)–(3) | |

Q4. Select the appropriate phrase for the blank marked as **(D)** which can make the given sentence contextually and grammatically correct.

- | | |
|---|---|
| (a) has flourished beyond all expectations due to global demand | (b) seem to operate under a cloud of regulatory scrutiny and talent drain |
|---|---|

(c) continues to generate unprecedented levels of foreign direct investment (d) remains insulated from the broader macroeconomic slowdown

(e) None of these

Q5. Select the appropriate filler for the blank marked as **(E)** in the given passage.

(i) Stagnation (ii) Proliferation (iii) Propensity

(a) Only (i)

(b) Only (iii)

(c) Both (i) and (iii)

(d) Both (ii) and (iii)

(e) None of these

Q6. In the above passage, a sentence **(F)** is given in italics. There may or may not be an error in one part of the sentence. Choose the part that has an error. If there is no error, choose (e).

(a) Confidence in independent arbiters,

(b) such as the judiciary, the free press, regulatory watchdogs

(c) and academic institutions,

(d) has been progressively hollowed out

(e) No error

Q7. The sentence given in **(G)** has four words given in bold. Which of the following must interchange to make the sentence grammatically and contextually correct and meaningful?

(a) (2)–(4)

(b) (1)–(2)

(c) (3)–(4)

(d) (1)–(3)

(e) (1)–(4)

Directions (8–12): Read the following passage and answer the questions. Some words are highlighted to help you answer some of the questions.

For generations, oceanographers held the conventional belief that the abyssal plains of the deep ocean were largely lifeless deserts — cold, dark and devoid of meaningful biological activity. This orthodoxy is now being overturned by researchers who have found thriving ecosystems clustered around hydrothermal vents on the ocean floor. These vents release superheated, mineral-rich water that sustains communities of chemosynthetic bacteria, which in turn support a remarkable food web of tube worms, clams, shrimp and even octopuses — all in the complete absence of sunlight.

It is well established that most surface life depends on photosynthesis for its primary energy. Marine biologist Dr. Rowena Solis reasoned that the chemosynthetic bacteria at hydrothermal vents harvest chemical energy from hydrogen sulphide rather than light energy from the sun. As the hot vent water rises and mixes with the frigid deep-sea water, it precipitates metallic sulphides that form towering chimney-like structures called 'black smokers,' _____ into the surrounding water column.

Once dismissed as anomalies in the 1970s, hydrothermal vent ecosystems gained scientific credibility when Dr. Solis and her team discovered an entirely new genus of tube worm near the Galápagos Rift — creatures that can grow to over two metres in length and live for more than two centuries. According to earlier assumptions, such longevity was impossible for animals living under such extreme pressure and temperature conditions. Dr. Solis, however, argued that the vent organisms had evolved novel enzymatic pathways that rendered the otherwise toxic hydrogen sulphide harmless and metabolically useful. This hypothesis was later confirmed by genomic sequencing, which revealed the presence of unique sulphur-binding proteins not found in any surface organism.

Q8. Which of the following is NOT in line with Dr. Solis's observations?

(a) Tube worms near the Galápagos Rift can survive for over two hundred years

(b) Chemosynthetic bacteria convert hydrogen sulphide into usable energy

(c) Black smokers release mineral-rich water that flows upward into the water column

(d) Vent organisms rely on photosynthetic pathways identical to surface plants

(e) None of these

Q9. Which of the following best defines the purpose of the passage?

(a) Detailing the commercial mining potential of hydrothermal vent minerals

(b) Challenging the conventional view that deep ocean floors are devoid of life

(c) Establishing the universality of photosynthesis across all ecosystems

(d) Both (b) and (c)

(e) None of these

Q10. Select the appropriate filler for the blank given in the passage.

- (a) Ascending
- (b) Dissolving
- (c) Dispersing minerals
- (d) Falling
- (e) None of these

Q11. Which of the following statement(s) is/are true with respect to the given passage?

- (a) The vent tube worms were already known to science before the 1970s
- (b) Chemosynthetic bacteria form the base of the food web at hydrothermal vents
- (c) Hydrogen sulphide is lethal to all known organisms
- (d) Both (a) and (b)
- (e) None of these

Q12. As per the passage, a 'black smoker' is best described as:

- (a) A deep-sea fish that absorbs carbon dioxide from vent water
- (b) A chimney-like mineral structure formed at hydrothermal vent sites
- (c) A type of chemosynthetic bacterium that oxidises sulphur compounds
- (d) A superheated plume of magma that rises from the ocean crust
- (e) None of these

Directions (13–19): Read the given passage carefully and answer the questions following the passage.

The global electric vehicle (EV) market accounted for roughly one fifth of all new car sales in 2023, with China alone absorbing more than half of those units. Yet the sheer scale of this market has not insulated incumbents from disruption. As recently as three years ago, legacy automakers from Germany, Japan and the United States dominated the premium segment. Then in 2021, a pair of Chinese brands — BYD and SAIC's MG — crashed into global consciousness and began eroding the market share of established names. By the end of that year, BYD had surpassed Tesla in total EV volumes globally, while MG became one of the top-selling EV brands in Europe and Southeast Asia. Just one year later, newer domestic brands such as NIO and XPeng further fragmented the premium EV segment, pushing legacy German marques toward the margins.

What happened to Tesla, briefly the pioneer and market leader? Its business model was built around vertical integration — from battery cells to software — and direct-to-consumer sales that bypassed traditional dealerships. Tesla sold its vehicles exclusively online and prided itself on industry-leading over-the-air software updates. However, this strategy later revealed a significant **vulnerability**: during the global semiconductor shortage of 2021–22, Tesla's reliance on proprietary chips meant it could not easily substitute components, leading to prolonged production delays and a shipment shortfall of several hundred thousand units, causing its market share to contract sharply.

In this market, the business model is decisive. BYD has cultivated deeply integrated partnerships with battery material suppliers across its entire supply chain, enabling it to maintain production continuity even during shortage cycles. Unlike Tesla or traditional automakers, BYD manufactures its own lithium iron phosphate (LFP) battery cells, which are cheaper, safer and longer-lasting than conventional nickel-manganese-cobalt cells. This gives BYD extraordinary cost control and pricing power.

BYD's distribution strategy also departs radically from convention. It targets not just wealthy city consumers but also the vast middle-market in Tier 3 and Tier 4 cities across China, supported by an extensive network of over 40,000 authorised service points. BYD backs this with aggressive sponsorships — from football clubs to entertainment franchises — to build brand salience in aspirational but price-conscious demographics. Finally, BYD shares attractive margins with its dealers, offering rebates of 6% to 12%, one of the highest in the industry, thereby securing fierce loyalty from its distribution network.

However, BYD's strategy of building _____(I)_____ manufacturing capacity carries inherent risks. Having more than doubled production in a single year, any significant slowdown in global EV adoption — driven by charging infrastructure gaps, policy reversals or battery recycling concerns — could leave BYD saddled with expensive idle capacity.

Q13. How has BYD's relationship with its supply chain partners benefited the company?

- (a) BYD entered markets after partnering exclusively with government-owned battery manufacturers
- (b) BYD secured uninterrupted production even during global component shortages
- (c) BYD successfully lobbied governments to impose tariffs on rival semiconductor makers
- (d) Both (b) and (c)
- (e) All of these

Q14. Which of the following has been BYD's approach to broadening its consumer base?

- (a) Targeting only premium urban consumers with flagship models priced above competitors
- (b) Focusing on Tier 3 and 4 city markets while avoiding expensive city-centre dealerships
- (c) Selling vehicles exclusively through digital platforms with no physical presence
- (d) BYD markets products through sponsorships appealing to aspirational demographics
- (e) Both (b) and (d)

Q15. How has the entry of new EV manufacturers affected the position of legacy automakers?

- (a) Established automakers retained stable market share due to superior brand loyalty
- (b) They responded by acquiring Chinese EV startups to diversify their portfolios
- (c) Newer brands captured market opportunities and shook the dominance of legacy names
- (d) Lower-priced EVs caused consumers to abandon premium brands for budget alternatives only
- (e) None of these

Q16. What were the consequences Tesla faced from its strategy of proprietary chip dependency?

- (i) The company lost bargaining power during supply shortages because it could not substitute components.
- (ii) Tesla experienced prolonged production delays and a significant shortfall in vehicle deliveries.
- (iii) Tesla was forced to recall all vehicles produced during 2021–22 due to software failures.
- (a) Only (ii)
- (b) Both (i) and (ii)
- (c) Both (ii) and (iii)
- (d) All (i), (ii) and (iii)
- (e) None of these

Q17. Which of the following has meaning opposite to **VULNERABILITY**, as highlighted in the passage?

- (a) Frailty
- (b) Resilience
- (c) Susceptibility
- (d) Quandary
- (e) Exposure

Q18. Which of the following words is similar in meaning to '**DECISIVE**' as used in the passage?

- (a) Ambiguous
- (b) Inconclusive
- (c) Pivotal
- (d) Peripheral
- (e) Tentative

Q19. Which of the following can fill in the blank (I) to make the statement grammatically and contextually correct? The word must also fit in the sentences given below:

- (A) The firm established a _____ trading platform to prevent unauthorised data sharing.
- (B) Investors were drawn to the company's _____ technology, which had no public documentation.

- (a) Incertitude
- (b) Proprietary
- (c) Duplicious
- (d) Collaborative
- (e) Qualm

Directions (20–24): In each of the questions below, four words are given in bold. These four words may or may not be in their correct position. Find the correct combination of words that should replace each other to make the sentence grammatically and contextually correct. If the sentence is already correct, select option (e).

Q20. The longer the (A)**standoff** persists, the (B)**tenser** diplomatic channels become (C)**stretched** beyond their limits, until the prospect of meaningful dialogue (D)**diminishes** entirely.

- (a) B–D
- (b) A–C
- (c) A–D
- (d) B–C
- (e) No replacement required

Q21. The hope is that once all (A)**posturing** is exhausted, pragmatic leaders will (B)**emerge** and guide both the disputed territory and the (C)**bilateral** relationship toward a framework where they can be (D)**resolved** collaboratively.

- (a) A–B and C–D
- (b) A–C and B–D
- (c) A–D
- (d) C–D
- (e) No replacement required

Q22. Civil (A)**rights** organisations and (B)**critical** voices of minority communities, who have been (C)**advocates** of Mr. Harmon, also condemned the administration for its erosion of (D)**groups**.

- (a) A–C
- (b) A–D and B–C
- (c) B–D and A–C
- (d) C–D
- (e) No replacement required

Q23. Despite what the summit **(A)achieved** for both delegations, a boundary may have been **(B)crossed** in conflating purely domestic **(C)partisan** interests with the nation's **(D)diplomacy**.

- (a) B–D
- (b) A–C
- (c) A–B
- (d) A–D
- (e) No replacement required

Q24. The minister has been **(A)praised** for having **(B)emerged** from his relative political **(C)inexperience** during the governance crisis, confronting vested interests and **(D)enabling** the ruling coalition to consolidate.

- (a) A–D and B–C
- (b) B–C
- (c) A–C and B–D
- (d) C–D
- (e) No replacement required

Directions (25–32): In the questions below, one part of the sentence is given in bold. The bold part may or may not be grammatically correct. Choose the best alternative to replace the bold part. If no replacement is needed, choose option (e) 'No replacement required'.

Q25. The scientist, a 34-year-old researcher from Bengaluru, **have been awarded on recognition** for her breakthrough in mRNA therapeutics.

- (a) have been awarded in recognition
- (b) has been awarded on recognition
- (c) has been awarded in recognition
- (d) has been awarded for recognising
- (e) No replacement required

Q26. Citizens will be eligible for subsidised housing and the municipal authority **can carried out** infrastructure upgrades in the identified zones.

- (a) could carried out
- (b) can carry forward
- (c) can carry out
- (d) carried out
- (e) No replacement required

Q27. **Resolving the ambiguity among** the two conflicting judgements and providing a definitive interpretation of Section 18, a seven-judge constitutional bench was assembled.

- (a) Resolving the ambiguity between
- (b) To resolve the ambiguity among
- (c) To resolve the ambiguity between
- (d) Resolution the ambiguity between
- (e) No replacement required

Q28. The city's air quality index reached the 'severe' category overnight, **and the gusty wind speed is keeping pollutants from dispersing further**.

- (a) but the gusty winds are keeping
- (b) and the gusty wind is keeping
- (c) and the gusty winds are keeping
- (d) and the gustier winds are keeping
- (e) No replacement required

Q29. Considering the latest meteorological projections available with SAFAR, air quality during the festival period **have been likely to be** marginally worse than the previous season.

- (a) is likely to be
- (b) have likely been
- (c) is likely to have been
- (d) is likely to had been
- (e) No replacement required

Q30. The upper atmosphere as well as surface-level boundary **layer winds is forecasted** to shift to a westerly pattern by Friday morning.

- (a) layer winds is forecast
- (b) layer winds are forecasting
- (c) layer winds has been forecasted
- (d) layer winds are forecast
- (e) No replacement required

Q31. Members of the student union and the faculty council jointly **held protests against the proposed** demolition of the century-old library building.

- (a) protest against the proposing
- (b) protested against the proposed
- (c) protests against the proposal
- (d) protest over the proposed
- (e) No replacement required

Q32. The administration **said they having received** multiple written complaints from faculty members alleging financial irregularities.

- (a) said they have received
(c) said it has received
(e) No replacement required
- (b) said they have been received
(d) said they had received

Directions (33–37): Read each of the following sentences to find out if there is any grammatical error. The error, if any, will be in one part of the sentence. The number of that part is your answer. If there is no error, mark (e).

Q33. I believe (A) / there are several researchers (B) / who would consider it (C) / inadvisable to proceed without further trials (D).

- (a) A
(c) C
(e) None of these
- (b) B
(d) D

Q34. Let us examine a theoretical (A) / scenario in which Priya (B) /, a recovering substance abuse patient (C) /, is offered a high-stress corporate role (D).

- (a) A
(c) C
(e) None of these
- (b) B
(d) D

Q35. Arjun and his colleague (A) / was travelling towards (B) / the conference venue when (C) / they encountered a familiar face (D).

- (a) A
(c) C
(e) None of these
- (b) B
(d) D

Q36. The resolution of the citizens' (A) / forum demanded that the prices of essential (B) / commodities be brought down (C) / and reviewed by the authority before the next quarter (D).

- (a) A
(c) C
(e) None of these
- (b) B
(d) D

Q37. The anxious official conducted (A) / a second thorough inspection (B) / of all departmental records, repeatedly (C) / including the annexures (D).

- (a) A
(c) C
(e) None of these
- (b) B
(d) D

Directions (38–40): Two columns are given containing three sentences/phrases each. Column 1 contains sentences A, B and C; Column 2 contains D, E and F. A sentence/phrase from Column 1 may or may not connect with one from Column 2 to form a grammatically and contextually correct sentence. Choose the pair(s) that make a grammatically meaningful sentence.

Q38. Column 1: (A) Commuters and rail passengers were advised to (B) The 2024 census is unlikely to capture 'caste-wise' (C) Minority rights groups staged protests **Column 2:** (D) occupational mobility across socio-economic strata (E) against the controversial citizenship amendment legislation (F) take alternate routes due to the flash flood warning

- (a) C–F and A–D
(c) A–F
(e) B–D and A–F
- (b) B–F
(d) C–E

Q39. Column 1: (A) Punjab National Bank, the country's second-largest public sector lender (B) A wave of structural reforms initiated by the government has led to (C) India has failed to fully capitalise on its services export potential **Column 2:** (D) significant formalisation of the unorganised workforce (E) challenging conditions for small and medium exporters (F) posted a net profit of ₹3,252 crore for the October–December quarter

- (a) C–F and A–D
(c) A–F
(e) B–D and A–F
- (b) B–F
(d) C–E

Q40. Column 1: (A) Bandhan Bank Ltd. has sought regulatory extension (B) The Securities and Exchange Board of India imposed a (C) The RBI has released fresh guidelines on **Column 2:** (D) the composite restructuring scheme for stressed borrowers (E) the tenure for which an independent director may serve on a bank's board (F) a penalty on twelve brokerages for

violating client fund segregation norms

- (a) C–F and A–D
- (b) B–F
- (c) A–F
- (d) C–E
- (e) B–D and A–F

Directions (41–45): The following sentences are not in their correct order. Rearrange them to form a coherent paragraph and answer the questions below.

(A) In the 2020–2022 period, of the 28 major States, only three — Kerala, Sikkim and Himachal Pradesh — had a Sex Ratio at Birth (SRB) above 950 girls per 1,000 boys.

(B) The practice of sex-selective abortion continues to rise at an alarming rate, according to the latest Sample Registration System data.

(C) Consequently, at present, an estimated 6% of girls are 'eliminated' before birth, notwithstanding the decade-long Beti Bachao Beti Padhao initiative.

(D) As a result, the Sex Ratio at Birth has been falling consistently since the 2011 Census.

(E) This figure has declined from 914 girls per thousand boys in 2015 to 899 girls in 2019.

Q41. Which of the following should be the **FIRST** sentence after rearrangement?

- (a) A
- (b) B
- (c) C
- (d) D
- (e) E

Q42. Which of the following should be the **FOURTH** sentence after rearrangement?

- (a) A
- (b) B
- (c) C
- (d) D
- (e) E

Q43. Which of the following should be the **SECOND** sentence after rearrangement?

- (a) A
- (b) B
- (c) C
- (d) D
- (e) E

Q44. Which of the following should be the **THIRD** sentence after rearrangement?

- (a) A
- (b) B
- (c) C
- (d) D
- (e) E

Q45. Which of the following should be the **LAST** sentence after rearrangement?

- (a) A
- (b) B
- (c) C
- (d) D
- (e) E

Directions (46–50): In each of the following questions, there are two statements with a missing word. Three words are given below each question. Choose the word that can fill both blanks to make meaningful sentences. If all three words are feasible, choose option (e) 'All (I), (II) and (III)'.

Q46. (1) The stark data immediately highlights why _____ transport infrastructure is so critical for economic integration.

(2) The supply networks formed by disruptive firms may not be as _____ as initially assumed, yet they continue to attract substantial venture capital.

(I) robust (II) expanding (III) resilient

- (a) Only (I)
- (b) Only (II)
- (c) Both (II) and (III)
- (d) Both (I) and (III)
- (e) All (I), (II) and (III)

Q47. (1) The data scientists developed a _____ algorithm that enabled them to track disinformation campaigns and study how narratives evolved post-election.

(2) According to trade publications, the logistics company Delhivery is set to launch its _____ intelligence suite called RouteIQ.

(I) predictive (II) tracking (III) outdated

- (a) Only (I)
- (b) Only (II)

- (c) Both (II) and (III)
(e) All (I), (II) and (III)

(d) Both (I) and (II)

Q48. (1) India's proposal for a greater leadership role for emerging economies in global health governance met with widespread _____.

(2) A coalition of environmental scientists and indigenous community leaders voiced their _____ to the plan to reroute the highway through protected forest land.

- (a) Only (I) opposition
(c) Both (II) and (III) support
(e) All (I), (II) and (III)

- (b) Only (II) gathering
(d) Both (I) and (III) opposition and support

Q48. (1) India's proposal for a greater leadership role in global health governance met with widespread _____.

(2) A coalition of environmental scientists voiced their _____ to the highway rerouting plan.

(I) opposition (II) gathering (III) support

- (a) Only (I)
(c) Both (II) and (III)
(e) All (I), (II) and (III)

- (b) Only (II)
(d) Both (I) and (III)

Q49. (1) In the _____ of our monsoon climate, every season breaks records in one direction or another.

(2) Temperature _____, erratic rainfall and prolonged droughts together account for up to 47% of variability in global food production.

(I) volatility (II) moderation (III) extremes

- (a) Only (I)
(c) Only (III)
(e) All (I), (II) and (III)

- (b) Only (II)
(d) Both (I) and (III)

Q50. (1) For the millions who wish to see the incumbent government _____, the upcoming state elections represent their most credible opportunity.

(2) One of the antique pillars was found _____ and lying at the base of the excavation trench.

(I) removed (II) dislodged (III) reinforced

- (a) Only (I)
(c) Both (II) and (III)
(e) All (I), (II) and (III)

- (b) Only (II)
(d) Both (I) and (II)

SECTION II: REASONING ABILITY (Q. 51–100)

Directions (51–55): Study the following information carefully and answer the questions.

Seven persons are living on different floors of a seven-storey building. The ground floor is numbered 1 and the topmost floor is numbered 7. They were each born in a different city.

A was born in Jaipur and lives on an even-numbered floor. G lives on the topmost floor. The number of persons living above A equals the number of persons living below F, who was born in Bhopal. D was born exactly two floors after the person born in Bhopal. C lives three floors above the person born in Surat. B was born in Nagpur. The person born in Indore lives above the person born in Lucknow.

Q51. Who among the following lives on the 5th floor?

- (a) C
- (b) The person born in Bhopal
- (c) D
- (d) The person born in Nagpur
- (e) Both (a) and (d)

Q52. How many persons live between the persons born in Jaipur and Surat?

- (a) Three
- (b) Four
- (c) None
- (d) Two
- (e) One

Q53. Four of the following five are alike in a certain way and form a group. Which does NOT belong to the group?

- (a) G – Jaipur
- (b) F – Surat
- (c) D – Surat
- (d) B – Lucknow
- (e) A – Indore

Q54. Who among the following lives on the bottommost floor?

- (a) B
- (b) The person born in Indore
- (c) The person born in Lucknow
- (d) F
- (e) E

Q55. A lives on the _____ floor.

- (a) 3rd
- (b) 4th
- (c) 2nd
- (d) 6th
- (e) None of these

Directions (56–59): A word and number arrangement machine rearranges words and numbers following a particular rule. Study the illustration below:

Input: 74 River Stone 56 Bright 38 Flame 91

Step I: 38 74 river stone 56 bright flame 91

Step II: 38 56 river stone bright flame 74 91

Step III: 38 56 bright river stone flame 74 91

Step IV: 38 56 bright flame river stone 74 91

Step V: 38 56 bright flame river stone 74 91 [Final Step]

Apply the same rules to the given input and answer the questions.

Input: 47 crown 19 throne 83 glory 62 nation

Q56. How many steps would be needed to complete the arrangement?

- (a) VI
- (b) VII
- (c) V
- (d) IV
- (e) None of these

Q57. Which step number would produce the following output?

'19 47 crown glory throne nation 62 83'

- (a) II
- (b) III
- (c) V
- (d) IV
- (e) None of these

Q58. Which of the following would be Step I?

- (a) 19 47 crown 62 throne glory nation 83
- (b) 19 62 crown glory throne nation 47 83

(c) 19 47 crown throne glory nation 62 83

(d) 19 47 crown 83 throne glory 62 nation

(e) None of the above

Q59. What is the sum of the numbers 3^{rd} from the left in Step III and 2^{nd} from the right in Step I?

(a) 109

(b) 145

(c) 130

(d) 126

(e) None of these

Directions (60–63): In each question, some statements are given followed by conclusions. Take the statements to be true even if they appear to contradict commonly known facts. Decide which conclusions logically follow.

Q60. Statements: Some Swift are Sedan. No Sedan is Hatchback. Only a few Hatchback are Compact.

Conclusions: I. All Compact being Sedan is a possibility. II. Some Swift is not Hatchback.

(a) Only I follows

(b) Both I and II follow

(c) Only II follows

(d) None follows

(e) None of these

Q61. Statements: Only Scarlet is Crimson. Some Indigo is Scarlet. Only a few Teal is Indigo.

Conclusions: I. Some Indigo being Crimson is a possibility. II. Some Scarlet being Teal is a possibility.

(a) Only I follows

(b) Both I and II follow

(c) Only II follows

(d) Either I or II follows

(e) None follows

Q62. Statements: Only a few Metropolis are Suburb. No Suburb is Township. Only a few Metropolis is Village.

Conclusions: I. Some Village is not Township. II. All Township is Metropolis.

(a) Only II follows

(b) Both I and II follow

(c) Only I follows

(d) Either I or II follows

(e) None follows

Q63. Statements: Only Firewall is Antivirus. All Encryption is Firewall. Few Patch is Encryption.

Conclusions: I. Some Encryption being Antivirus is a possibility. II. Some Firewall being Patch is a possibility.

(a) Only I follows

(b) Both I and II follow

(c) Only II follows

(d) Either I or II follows

(e) None follows

Directions (64–65): Each question below has two statements numbered I and II. Decide whether the data in the statements are sufficient to answer the question.

(a) Statement I alone is sufficient, but II alone is not. (b) Statement II alone is sufficient, but I alone is not.

(c) Either statement alone is sufficient. (d) Neither statement, even together, is sufficient.

(e) Both statements together are necessary and sufficient.

Q64. What does 'ke' mean in the coded language?

I. 'ma ke ri' means 'Buy fresh apples' and 'ke su ma ri' means 'Never buy stale apples' in that code.

II. 'pe ve ke ra' means 'Buy now or wait' and 'zo wx si ke' means 'Either buy or pass' in that code.

(a) Statement I alone

(b) Statement II alone

(c) Either alone

(d) Neither alone nor together

(e) Both together necessary

Q65. Point M is in which direction with respect to point N?

I. Point P is west of point Q and point N is south of point P. Point M is north-east of point P.

II. Point N is east of point Q and south of point P. Point M is north of point P.

(a) Statement I alone

(b) Statement II alone

(c) Either alone

(d) Neither alone nor together

(e) Both together necessary

Directions (66–70): Study the information carefully and answer the questions below.

Nine boxes are arranged in a single stack (one above another). Not more than five boxes are placed above box P.

Four boxes are placed between box P and box Q. Box E is placed immediately above box J. The number of boxes

below Q equals the number of boxes above box R. Three boxes are placed between box S and box R (S is above R).

Box P is not at the topmost position. Four boxes are placed between box E and box T (T is above E). The number of boxes between box T and box U equals the number between box T and box S. Box V does not sit below box J.

Q66. How many boxes are placed between box V and box T?

- (a) Three
- (b) More than Three
- (c) One
- (d) Two
- (e) None of these

Q67. Which of the following box is placed just above box S?

- (a) T
- (b) U
- (c) J
- (d) V
- (e) None of these

Q68. Which of the following statements is true about box E?

- (a) Four boxes are placed between box E and box U
- (b) Box E is at the topmost position
- (c) Box E is placed third from the bottommost position
- (d) Box S is placed immediately above box E
- (e) None is true

Q69. Which of the following box is placed exactly in the middle of the stack?

- (a) S
- (b) T
- (c) P
- (d) Q
- (e) None of these

Q70. Which of the following box is placed immediately below box Q?

- (a) T
- (b) V
- (c) J
- (d) E
- (e) None

Directions (71–74): These questions are based on the following arrangement. Study it carefully to answer the questions.

% B F K M P @ Q R # 6 3 D G 5 S H 7 ? J C 8 W L N

Q71. How many such symbols are there in the above arrangement, each of which is immediately followed by a number and also immediately preceded by a consonant?

- (a) None
- (b) One
- (c) Two
- (d) Three
- (e) None of these

Q72. If we remove all the vowels from this series, which of the following element will be 7th from the right end?

- (a) S
- (b) H
- (c) D
- (d) Q
- (e) None of these

Q73. Which of the following element is 3rd to the right of the element that is 7th from the left end?

- (a) #
- (b) R
- (c) H
- (d) *
- (e) None of these

Q74. How many such consonants are there in the above arrangement, each of which is immediately preceded by a consonant and also immediately followed by a consonant?

- (a) None
- (b) One
- (c) Two
- (d) Three
- (e) More than three

Q75. Digital financial literacy among rural households in India remains critically low, with over 60% of surveyed respondents unable to identify phishing attempts or verify the authenticity of UPI payment requests. Fraudulent transactions targeting first-generation digital users have risen by 38% year-on-year.

What can be the course of action to improve financial safety for rural digital users?

(I) Banks and fintech firms should deploy local-language cybersecurity awareness campaigns in rural areas via community radio

and village-level self-help groups.

(II) The government should mandate that all UPI transactions above ₹500 require fingerprint biometric verification in rural districts.

(III) Rural households should be encouraged to avoid digital transactions entirely and revert to cash-based economies.

(IV) Gram panchayat digital kiosks should be equipped with transaction-verification helplines staffed by trained banking correspondents.

- (a) Only I and IV
(b) Only II, III and IV
(c) Only III and IV
(d) Only I, II and IV
(e) All of them

Q76. The Union Health Ministry announced that a new drug-resistant strain of tuberculosis has been detected in six States, with the highest concentration in urban slum clusters. The Chief Medical Officer stated: 'The most critical intervention is to ensure complete treatment adherence. Incomplete courses are the primary driver of drug resistance. Case detection and contact tracing will limit community spread.'

Which of the following can be inferred from the above statement?

(I) The government is planning a nationwide hospital bed expansion to treat all TB cases.

(II) The number of active case-detection drives will likely increase in the affected States.

(III) Drug-resistant TB in urban slum clusters is a matter of serious public health concern.

- (a) Only I and III
(b) Only II
(c) Only I
(d) Only II and III
(e) All three

Directions (77–79): Study the following information and answer the questions.

There are eight members in a family with three married couples across three generations and an equal number of males and females. X is the father of Y. W is the father of T. Y is the sister of Z. S is the daughter of V. S is the sister of T. U is the son-in-law of V. T has two children.

Q77. Who among the following is the mother of Z?

- (a) T
(b) S
(c) V
(d) Cannot be determined
(e) None of these

Q78. How is V related to Y?

- (a) Grandfather
(b) Mother
(c) Grandmother
(d) Brother
(e) None of these

Q79. Who among the following is the brother-in-law of S?

- (a) W
(b) U
(c) Y
(d) X
(e) None of these

Directions (80–84): Study the information carefully and answer the questions below.

Eight persons are sitting around a circular table. Some face the centre; others face outward.

L sits third to the right of Q. Only one person sits between L and R. Four persons sit between R and K. M sits second to the right of K. N sits second to the right of O. N does not sit adjacent to L. Immediate neighbours of K face the opposite direction. P sits immediately right of N. Immediate neighbours of Q face the opposite direction. O sits immediately right of L. O faces P. P faces the opposite direction of R.

Q80. Who among the following sits third to the right of R?

- (a) O
(b) M
(c) P
(d) N
(e) None of these

Q81. Who among the following faces M?

- (a) R
(b) O
(c) L
(d) K

(e) None of these

Q82. How many persons sit between N and K, when counted to the right of N?

- (a) One
- (b) Two
- (c) Three
- (d) Four
- (e) None of these

Q83. Which of the following statements is true about L?

- (a) L faces the same direction as K
- (b) N sits opposite to L
- (c) L sits immediately right of R
- (d) M sits second to the right of L
- (e) None is true

Q84. Who among the following sits immediately right of the person who sits third to the left of P?

- (a) K
- (b) Q
- (c) M
- (d) N
- (e) None of these

Directions (85–87): Study the information carefully and answer the questions below.

Five aircraft perform an aerial exercise. Alpha is 12 km north of Bravo. Port Z is exactly midway between Alpha and Bravo. Charlie is 9 km east of Bravo. Delta is 9 km east of Port Z. Echo is 5 km north of Delta.

Q85. What is the shortest distance between Echo and Charlie?

- (a) 12 km
- (b) 10 km
- (c) 8 km
- (d) 7 km
- (e) 14 km

Q86. If Charlie moves 18 km to the west, what is the direction of Delta with respect to Charlie?

- (a) North-east
- (b) South-west
- (c) North-west
- (d) South-east
- (e) North

Q87. What is the shortest distance between Echo and Port Z?

- (a) 12 km
- (b) 14 km
- (c) 9 km
- (d) 7 km
- (e) None of these

Q88. It is widely accepted that students retain information more effectively when learning is contextualised within real-world scenarios rather than delivered through abstract lectures. Educational researchers observed that learners who were exposed to case-study-based instruction scored 34% higher on retention assessments a fortnight later compared to those taught through traditional didactic methods.

Which of the following weakens the argument above?

- (a) Several leading universities have adopted case-study curricula across engineering programmes.
- (b) The study was conducted exclusively on postgraduate students aged 24–30.
- (c) Abstract instruction methods have been deprecated by the National Curriculum Framework.
- (d) Case studies used in the research were sourced from industries unrelated to the students' fields.
- (e) Both (b) and (d)

Directions (89–90): Each question has two statements I and II.

- (a) Statement I alone is sufficient
- (b) Statement II alone is sufficient
- (c) Either alone is sufficient
- (d) Neither alone nor together is sufficient
- (e) Both together are necessary and sufficient

Q89. In a row, what is the total number of persons?

- I. Divya's position is 18th from the left end and Sameer's position is 26th from the right end.
- II. Prachi is 20th from the left end and 17th from the right end.

- (a) Statement I alone
- (b) Statement II alone
- (c) Either alone
- (d) Neither alone nor together
- (e) Both together

Q90. Among five colleagues A, B, C, D and E, who earns the most, if all salaries differ?

I. B earns more than D. C earns more than B and E. D is not the lowest earner.

II. A earns more than C but is not the highest earner.

- | | |
|-----------------------|--------------------------------|
| (a) Statement I alone | (b) Statement II alone |
| (c) Either alone | (d) Neither alone nor together |
| (e) Both together | |

Directions (91–95): Study the following information carefully and answer the questions.

Eight persons sit in a row. Four face North; four face South. Only three persons sit between A and B. A sits at one of the extreme ends. C sits third to the left of B and faces the same direction as B. More than two persons sit between D and E. F sits third to the left of G. H sits second to the left of E; both face opposite directions. Persons at the extreme ends face opposite directions. C does not sit at an extreme end. Neither D nor E is an immediate neighbour of B. G faces the opposite direction to A and D. F does not face North and is not an immediate neighbour of H.

Q91. Who among the following sits exactly between C and the person who sits second to the right of H?

- | | |
|-------|-------|
| (a) A | (b) D |
| (c) F | (d) E |
| (e) G | |

Q92. Who among the following sits third to the left of F?

- | | |
|-------|-------|
| (a) A | (b) G |
| (c) B | (d) C |
| (e) H | |

Q93. Four of the following five are alike in a certain way and form a group. Who does NOT belong to the group?

- | | |
|-------|-------|
| (a) A | (b) D |
| (c) B | (d) F |
| (e) G | |

Q94. How many people sit between A and H?

- | | |
|----------|-----------|
| (a) Four | (b) Three |
| (c) None | (d) Two |
| (e) One | |

Q95. Who among the following sits fourth to the right of the person who sits second to the left of E?

- | | |
|-------|-------|
| (a) H | (b) G |
| (c) D | (d) C |
| (e) F | |

Directions (96–99): Relationships between elements are shown in statements. Decide which conclusions follow.

(a) Only conclusion I follows (b) Only conclusion II follows

(c) Either I or II follows (d) Neither follows (e) Both follow

Q96. Statements: $X1 > X2 = X3 < X4 \leq X5 > X6$

Conclusions: I. $X5 > X2$ II. $X6 > X3$

- | | |
|--------------------|-------------|
| (a) Only I | (b) Only II |
| (c) Either I or II | (d) Neither |
| (e) Both I and II | |

Q97. Statements: $A \geq B > C \geq D < E \leq F$

Conclusions: I. $D < A$ II. $D < F$

- | | |
|--------------------|-------------|
| (a) Only I | (b) Only II |
| (c) Either I or II | (d) Neither |
| (e) Both I and II | |

Q98. Statements: $P \leq Q < R > S = T \geq U$

Conclusions: I. $Q \geq U$ II. $U \leq S$

- | | |
|--------------------|-------------|
| (a) Only I | (b) Only II |
| (c) Either I or II | (d) Neither |

(e) Both I and II

Q99. Statements: $M = N \geq O \geq P < Q > R$

Conclusions: I. $O < P$ II. $R > N$

(a) Only I

(b) Only II

(c) Either I or II

(d) Neither

(e) Both I and II

Q100. If in the number '7482516390', positions of the 1st and 2nd digits are interchanged, the 3rd and 4th digits are interchanged, and so on until the 9th and 10th digits are interchanged, which digit will be 5th from the left end?

(a) 5

(b) 1

(c) 6

(d) 2

(e) None of these

SECTION III: QUANTITATIVE APTITUDE (Q. 101–150)

Directions (101–106): The table below shows six institutes with five programmes each — Engineering, Management, Law, Science and Arts — along with total applications received, percentage of cancelled applications and the male-to-female ratio among approved applicants.

Programme	Total Applications	% Cancelled	M : F (Approved)
Engineering	P	45%	13 : 7
Management	1600	Q	31 : 19
Law	1800	40%	R
Science	S	60%	47 : 13
Arts	900	65%	T

Q101. If the difference between approved male and female applications in Engineering is 240, find how much percent P is more than total applications in the Arts programme.

- (a) 75% (b) 87.5%
(c) 62.5% (d) 66%
(e) 50%

Q102. If the difference between approved male and female applicants in Science is 272, find the sum of total applications received in Science, Management and Law together.

- (a) 4200 (b) 4800
(c) 5000 (d) 4400
(e) 5200

Q103. If the difference between approved male and female applicants in Engineering is 120, find the difference between P and the number of cancelled applications in Law.

- (a) 170 (b) 180
(c) 190 (d) 165
(e) 175

Q104. If the number of approved male applicants in Arts exceeds approved female applicants by 35, find the ratio of approved female to approved male applicants in Arts.

- (a) 9 : 7 (b) 5 : 7
(c) 13 : 9 (d) 7 : 9
(e) 7 : 11

Q105. Find the ratio of cancelled applications in Law to those in Arts.

- (a) 7 : 5 (b) 5 : 7
(c) 8 : 7 (d) 9 : 7
(e) 7 : 6

Q106. If the difference between approved male and female applicants in Management is 136, find the sum of Q and total applications in Law.

- (a) 2300 (b) 2450
(c) 2700 (d) 2850
(e) 3200

Q107. Pipe X alone can fill a cistern in 24 mins and pipe Y alone can fill it in 'N' mins. Pipe Z alone can empty it in 48 mins. If pipes X, Y and Z together take $18\frac{6}{7}$ min to fill the cistern, find the time taken by pipe Y alone to fill $\frac{4}{5}$ of the cistern.

- (a) 12 min (b) 18 min
(c) 24 min (d) 28 min
(e) 32 min

Q108. A container holds 280 litres of a tin-copper alloy of which 42.5% is copper. 56 litres of the mixture are removed and 22 litres of copper and 16 litres of tin are added. Find the ratio of copper to tin in the final mixture.

- (a) 47 : 31 (b) 47 : 29
(c) 29 : 47 (d) 43 : 31
(e) 31 : 47

Q109. Train A of length 540 m crosses a platform that is twice its length in 48 sec. Train B running at 63 kmph crosses a stationary pole in 40 sec. Find the time taken by both trains to cross each other when running in the same direction.

- (a) $28\frac{2}{7}$ sec (b) $24\frac{2}{7}$ sec
(c) $34\frac{2}{7}$ sec (d) $36\frac{2}{7}$ sec
(e) $30\frac{2}{7}$ sec

Q110. Preeti and Rajat started a venture with capitals of Rs 4,200 and Rs 5,600 respectively. After five months Preeti increased her investment by 14.28% and Rajat withdrew $8\frac{1}{9}\%$ from his investment. If at year-end the difference in their profit shares is Rs 1,960, find Rajat's profit share.

- (a) Rs 10,850 (b) Rs 10,920
(c) Rs 11,200 (d) Rs 10,080
(e) Rs 11,760

Q111. The age of Kavya is 20% more than the age of Shreya. The ratio of Shreya's present age to Rohan's present age is 3 : 7, and after 10 years the average age of Rohan and Kavya will be 38 years. Find the sum of Kavya's age 5 years hence and Rohan's age 4 years ago.

- (a) 54 yrs (b) 48 yrs
(c) 52 yrs (d) 46 yrs
(e) 50 yrs

Q112. If the difference between total compound interest over two years and total simple interest over three years on a certain principal at 10% p.a. is Rs 3,100, what will be the compound interest earned when three times the same principal is invested at 8% p.a. for two years?

- (a) Rs 12,480 (b) Rs 16,800
(c) Rs 15,600 (d) Rs 17,800
(e) Rs 19,200

Q113. If the length of a rectangle is increased by 3 cm it becomes the side of a square. The area of the square is 30 cm^2 more than the area of the rectangle and the ratio of length to breadth of the rectangle is 7 : 6. Find the ratio of the area of the rectangle to the area of the square.

- (a) 3 : 4 (b) 5 : 6
(c) 14 : 15 (d) 6 : 7
(e) None of these

Q114. P invested Rs M at 11% p.a. and Q invested Rs (M – 5,000) at 13% p.a. If both receive total compound interest of Rs 7,816.53 at the end of two years, find the capital invested by Q.

- (a) Rs 16,000 (b) Rs 12,000
(c) Rs 10,000 (d) Rs 14,000
(e) Rs 18,000

Q115. The ratio of the present salaries of Anand and Prakash is 7 : 6, and the ratio of their current salary to last year's salary is 5 : 4 and 4 : 3 respectively. If the difference between their previous year's salaries is Rs 2,500, find Prakash's current salary.

- (a) Rs 24,000 (b) Rs 18,000
(c) Rs 20,000 (d) Rs 16,000
(e) Rs 21,000

Q116. A retailer offers a 25% discount on each article and also gives one free article for every five purchased, yet earns a 10% profit. If the MRP of each article is Rs 320, find the cost price of one article when he buys six articles.

- (a) Rs 150 (b) Rs 120
(c) Rs 160 (d) Rs 140
(e) None of these

Q117. Y men can complete a project in 18 days working 9 hours each day. The number of days taken by Y men is 75% of the number of days taken by (Y – 1) men working 7 hours a day. Find the value of Y.

- (a) 14 (b) 13
(c) 15 (d) 16

(e) 17

Directions (118–122): What will come in place of the question mark (?) in the following number series?

Q118. 14, 9, 10.5, 17.25, ?, 91

- | | |
|--------|--------|
| (a) 34 | (b) 28 |
| (c) 32 | (d) 36 |
| (e) 42 | |

Q119. 15, 84, 430, 1724, 5175, ?

- | | |
|-----------|-----------|
| (a) 10352 | (b) 10356 |
| (c) 10358 | (d) 10362 |
| (e) 10420 | |

Q120. 31, 64, 258, 1549, 12396, ?

- | | |
|------------|------------|
| (a) 111565 | (b) 111569 |
| (c) 112000 | (d) 111005 |
| (e) 113005 | |

Q121. 5218, 3576, 2668, 2214, 2060, ?

- | | |
|----------|----------|
| (a) 2012 | (b) 2018 |
| (c) 2005 | (d) 2008 |
| (e) 2022 | |

Q122. 32, 32, 64, 21.33, 85.33, ?

- | | |
|----------|----------|
| (a) 21.3 | (b) 17.1 |
| (c) 19.5 | (d) 13.4 |
| (e) 15.6 | |

Directions (123–127): The pie charts below show the percentage distribution of five occupational categories in two districts — District X and District Y. Read the data carefully and answer the questions.

District X: Doctors 12% | Engineers 22% | Traders 28% | Lawyers 24% | Teachers 14%

District Y: Doctors 18% | Engineers 27% | Traders 28% | Lawyers 18% | Teachers 9%

Q123. If total professionals in District Y is 35% more than in District X, find the approximate combined percentage of Engineers across District X and District Y together.

- | | |
|---------|---------|
| (a) 19% | (b) 25% |
| (c) 21% | (d) 23% |
| (e) 17% | |

Q124. Total Teachers in both districts is 116 and total Traders in District X is 42 more than total Traders in District Y. Find total Lawyers in both districts.

- | | |
|---------|---------|
| (a) 184 | (b) 198 |
| (c) 210 | (d) 192 |
| (e) 204 | |

Q125. Total Doctors in District X is $88\frac{8}{9}\%$ of total Doctors in District Y. Find by what percent total Engineers in District X exceed total Engineers in District Y.

- | | |
|--------|---------|
| (a) 1% | (b) 3% |
| (c) 5% | (d) 6■% |
| (e) 2% | |

Q126. The ratio of total professionals in District X to District Y is 9 : 8. The difference between total Doctors in District X and total Engineers in District Y is 126. Find the total number of Traders in both districts.

- | | |
|---------|---------|
| (a) 540 | (b) 560 |
| (c) 480 | (d) 504 |
| (e) 520 | |

Q127. Total professionals in District Y is 25% more than in District X. Find the ratio of total Lawyers in District X to total Engineers in District Y.

- (a) 5 : 4
(c) 4 : 5
(e) 5 : 3
- (b) 5 : 6
(d) 5 : 9

Q128. The average of five numbers is 72. If 4 is added to the first, multiplied by the second, subtracted from the third and divided into the fourth and fifth, all resulting values are equal. Find the difference between the 2nd highest and the smallest number.

- (a) 45
(c) 42
(e) 32
- (b) 35
(d) 39

Q129. 5 years ago, the ratio of ages of P and Q was 4 : 1. If the present age of Q is 37.5% of the present age of P, what is the difference between their present ages?

- (a) 25
(c) 35
(e) 45
- (b) 30
(d) 40

Q130. The ratio of the volume of a conical tent to the total surface area of a hemispherical bowl is 6 : 1. The radius of the bowl and the cone are equal and the slant height of the cone is 90 cm. If the radius of the cone equals the side of a cube, find the total surface area of the cube.

- (a) 1944 cm²
(c) 2916 cm²
(e) 1512 cm²
- (b) 2160 cm²
(d) 1690 cm²

Q131. 'X' purchased two motorcycles at the same price. X sold one to 'Y' at 55% profit and the other to 'Z' at 25% profit. 'Z' sold the motorcycle to 'W' at 15% less than the average cost price paid by Y and Z. If W incurred a 20% loss by selling it to 'V' for Rs 2,520, find the difference between the cost prices paid by X and W.

- (a) Rs 700
(c) Rs 900
(e) Rs 600
- (b) Rs 450
(d) Rs 300

Q132. Speed of a boat downstream is 2.5 times the speed upstream. If the boat covers 112 km downstream and 84 km upstream in a total of 22 hours, find the still water speed of the boat.

- (a) 6 km/hr
(c) 7 km/hr
(e) 12 km/hr
- (b) 8 km/hr
(d) 9 km/hr

Q133. A bag contains some red balls and a total of 11 black and yellow balls. The probability of drawing two red balls is $\frac{2}{11}$, which is 300% of the probability of drawing two black balls. Find the number of yellow balls if the number of black balls is even.

- (a) 3
(c) 6
(e) None of these
- (b) 5
(d) 4

Q134. In a college election between two candidates, 75% of students cast their votes, of which 150 votes were declared invalid. The winner received 40% of the total enrolled students' votes and won by exactly 45 votes. Find the number of students who did not vote.

- (a) 360
(c) 340
(e) 380
- (b) 320
(d) 300

Q135. Shweta drives from Pune to Nashik at 90 kmph. After 135 km she reduces her speed by 10% and completes the entire journey in 6 hours. Find the total time taken if she had reduced her speed after 2.5 hours instead.

- (a) 6 hours
(c) $5\frac{3}{7}$ hours
(e) 4.8 hours
- (b) 5.8 hours
(d) $4\frac{3}{7}$ hours

Directions (136–140): The line graph below shows the number of tablets (Lenovo and Samsung) sold by a retailer over five days (Mon–Fri). Read the graph carefully and answer the questions.

Lenovo: Mon 85, Tue 100, Wed 60, Thu 55, Fri 105

Samsung: Mon 75, Tue 120, Wed 42, Thu 80, Fri 90

Q136. What is the difference between the average number of Lenovo tablets sold on Wednesday, Thursday and Friday and the average Samsung tablets sold on Tuesday, Wednesday and Thursday?

- (a) 20
- (b) 10
- (c) 15
- (d) 5
- (e) 25

Q137. If the price of a Lenovo tablet is 15% more than a Samsung tablet, the revenue earned on Tuesday is what percentage of the revenue earned on Thursday?

- (a) 60%
- (b) 75%
- (c) 50%
- (d) 65%
- (e) 80%

Q138. Total tablets sold on Wednesday is what percentage more/less than total tablets sold on Monday?

- (a) 20%
- (b) 40%
- (c) 35%
- (d) 50%
- (e) 25%

Q139. From Monday to Friday, 15% of total Samsung tablets were found defective and 55% of the remaining were unsold. Find the total number of Samsung tablets sold (non-defective).

- (a) 185
- (b) 163
- (c) 172
- (d) 154
- (e) 195

Q140. If the ratio of Lenovo to Apple tablets sold by the retailer on Wednesday, Thursday and Friday is 3:5, 4:3 and 5:6 respectively, find the total Apple tablets sold on these three days.

- (a) 240
- (b) 220
- (c) 255
- (d) 250
- (e) 265

Directions (141–145): Read the information carefully and answer the questions.

A, B, C and D started a business in partnership. A invested Rs 6,000 for the whole year and the average initial investment of all four partners is Rs 6,975. D's investment is 30% more than B's, and the ratio of B's to C's investment is 10 : 9. Time for which B invested is $\frac{3}{4}$ of A's investment period. The sum of investment periods of C and D equals the total investment period of A and B. The profit share of C is $\frac{18}{79}$ of total profit at year-end.

Q141. Find the ratio of the investment period of C to that of D.

- (a) 1 : 1
- (b) 1 : 2
- (c) 2 : 3
- (d) 3 : 2
- (e) 4 : 3

Q142. The difference between D's and C's investment is what percent of the difference between B's and D's investment?

- (a) 125%
- (b) 100%
- (c) 150%
- (d) 140%
- (e) 160%

Q143. If the total profit at year-end is Rs 19,025, find the difference between the profit shares of A and D.

- (a) Rs 800
- (b) Rs 1,200
- (c) Rs 1,000
- (d) Rs 1,500
- (e) None of these

Q144. B's profit share is what percent of A's profit share?

- (a) 80%
- (b) 75%
- (c) 60%
- (d) 100%
- (e) 65%

Q145. The combined investment of C and D is approximately what percent of the total investment of all four partners?

- (a) 45%
 - (b) 48%
 - (c) 50%
 - (d) 52%
 - (e) 54%
-

Directions (146–150): Find the approximate value that should come in place of the question mark (?) in each question. (Exact calculation is not required.)

Q146. $?^2 + 224.97 - 21.02 \times 7.98 = 383.01 \times 1.99$

- | | |
|--------|--------|
| (a) 14 | (b) 19 |
| (c) 26 | (d) 28 |
| (e) 22 | |

Q147. $864.08 \div 11.98 \times \sqrt{289.04} - 64.05 = ?$

- | | |
|----------|----------|
| (a) 1160 | (b) 1225 |
| (c) 980 | (d) 1100 |
| (e) 1040 | |

Q148. $38.01\% \text{ of } 600.09 \div ? = 342.07 \div \sqrt{784.95}$

- | | |
|--------|--------|
| (a) 14 | (b) 21 |
| (c) 9 | (d) 28 |
| (e) 17 | |

Q149. $(? \div 11.98) \times 14.03 = 22.01\% \text{ of } 1499.97$

- | | |
|---------|---------|
| (a) 220 | (b) 260 |
| (c) 300 | (d) 280 |
| (e) 240 | |

Q150. $? \% \text{ of } 199.99 = \sqrt{((27.01)^2 + (19.98)^2)} + 65.01\% \text{ of } 699.98$

- | | |
|--------|--------|
| (a) 80 | (b) 60 |
| (c) 40 | (d) 20 |
| (e) 10 | |